

IIT JAM 2007 Official Paper

Category: Classic Era (2005–2014) • Official Mock Test Series

TOTAL QUESTIONS

15

TOTAL MARKS

90 M

TEST DURATION

60 Min

PAPER PATTERN

MCQ

Section / Type		Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)		15	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2007 • Q1 • ID: JAM_2007_Q1)

MCQ

+6 M

Neg: -2.0

1. Let $A(t)$ denote the area bounded by the curve $y=e^{-|x|}$, the x -axis and the straight lines $x=-t$ and $x=t$. Then $\lim_{t \rightarrow \infty} A(t)$ is equal to
- (A) 2
(B) 1
(C) $1/2$
(D) 0

Question 2 (JAM 2007 • Q2 • ID: JAM_2007_Q2)

MCQ

+6 M

Neg: -2.0

2. If k is a constant such that $xy+k=e^{(x-1)^2/2}$ satisfies the differential equation $x \frac{dy}{dx} = (x^2 - x - 1)y + (x - 1)$, then k is equal to
- (A) 1
(B) 0
(C) -1
(D) -2

Question 3 (JAM 2007 • Q3 • ID: JAM_2007_Q3)

MCQ

+6 M

Neg: -2.0

3. Which of the following functions is uniformly continuous on the domain as stated?
- (A) $f(x) = x^2, \quad x \in \mathbf{R}$
(B) $f(x) = \frac{1}{x}, \quad x \in [1, \infty)$
(C) $f(x) = \tan x, \quad x \in (-\pi/2, \pi/2)$
(D) $f(x) = [x], \quad x \in [0, 1]$
([x] is the greatest integer less than or equal to x)

Question 4 (JAM 2007 • Q4 • ID: JAM_2007_Q4)

MCQ

+6 M

Neg: -2.0

4. Let R be the ring of polynomials over $\mathbb{Z}_2$ and let I be the ideal of R generated by the polynomial $x^3 + x + 1$. Then the number of elements in the quotient ring R/I is

- (A) 2
- (B) 4
- (C) 8
- (D) 16

Question 5 (JAM 2007 • Q5 • ID: JAM_2007_Q5)

MCQ

+6 M

Neg: -2.0

5. Which of the following sets is a basis for the subspace

$$W = \left\{ \begin{bmatrix} x & y \\ 0 & t \end{bmatrix} : x + 2y + t = 0, y + t = 0 \right\}$$

of the vector space of all real 2×2 matrices?

- (A) $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$
- (B) $\left\{ \begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \right\}$
- (C) $\left\{ \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix} \right\}$
- (D) $\left\{ \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \right\}$

Question 6 (JAM 2007 • Q6 • ID: JAM_2007_Q6)

MCQ

+6 M

Neg: -2.0

6. Let G be an Abelian group of order 10. Let $S = \{g \in G : g^{-1} = g\}$. Then the number of non-identity elements in S is

- (A) 5
- (B) 2
- (C) 1
- (D) 0

Question 7 (JAM 2007 • Q7 • ID: JAM_2007_Q7)

MCQ

+6 M

Neg: -2.0

7. Let (a_n) be an increasing sequence of positive real numbers such that the series $\sum_{k=1}^{\infty} a_k$ is divergent. Let $s_n = \sum_{k=1}^n a_k$ for $n = 1, 2, \dots$ and $t_n = \sum_{k=2}^n \frac{a_k}{s_{k-1}s_k}$ for $n = 2, 3, \dots$. Then $\lim_{n \rightarrow \infty} t_n$ is equal to

- (A) $1/a_1$
 (B) 0
 (C) $1/(a_1 + a_2)$
 (D) $a_1 + a_2$

Question 8 (JAM 2007 • Q8 • ID: JAM_2007_Q8)

MCQ

+6 M

Neg: -2.0

8. For every function $f: [0,1] \rightarrow \mathbf{R}$ which is twice differentiable and satisfies $f'(x) \geq 1$ for all $x \in [0,1]$, we must have
- (A) $f''(x) \geq 0$ for all $x \in [0,1]$
 (B) $f(x) \geq x$ for all $x \in [0,1]$
 (C) $f(x_2) - x_2 \leq f(x_1) - x_1$ for all $x_1, x_2 \in [0,1]$ with $x_2 \geq x_1$
 (D) $f(x_2) - x_2 \geq f(x_1) - x_1$ for all $x_1, x_2 \in [0,1]$ with $x_2 \geq x_1$

Question 9 (JAM 2007 • Q9 • ID: JAM_2007_Q9)

MCQ

+6 M

Neg: -2.0

9. Let $f: \mathbf{R}^2 \rightarrow \mathbf{R}$ be defined by

$$f(x,y) = \begin{cases} \frac{x^2 y}{x^4 + y^2} & \text{if } (x,y) \neq (0,0), \\ 0 & \text{if } (x,y) = (0,0). \end{cases}$$

Which of the following statements holds regarding the continuity and the existence of partial derivatives of f at $(0,0)$?

- (A) Both partial derivatives of f exist at $(0,0)$ and f is continuous at $(0,0)$
 (B) Both partial derivatives of f exist at $(0,0)$ and f is NOT continuous at $(0,0)$
 (C) One partial derivative of f does NOT exist at $(0,0)$ and f is continuous at $(0,0)$
 (D) One partial derivative of f does NOT exist at $(0,0)$ and f is NOT continuous at $(0,0)$

Question 10 (JAM 2007 • Q10 • ID: JAM_2007_Q10)

MCQ

+6 M

Neg: -2.0

10. Suppose (c_n) is a sequence of real numbers such that $\lim_{n \rightarrow \infty} |c_n|^{1/n}$ exists and is non-zero.

If the radius of convergence of the power series $\sum_{n=0}^{\infty} c_n x^n$ is equal to r , then the radius of convergence of the power series $\sum_{n=1}^{\infty} n^2 c_n x^n$ is

- (A) less than r
- (B) greater than r
- (C) equal to r
- (D) equal to 0

Question 11 (JAM 2007 • Q11 • ID: JAM_2007_Q11)

MCQ

+6 M

Neg: -2.0

11. The rank of the matrix $\begin{bmatrix} 1 & 4 & 8 \\ 2 & 10 & 22 \\ 0 & 4 & 12 \end{bmatrix}$ is

- (A) 3
- (B) 2
- (C) 1
- (D) 0

Question 12 (JAM 2007 • Q12 • ID: JAM_2007_Q12)

MCQ

+6 M

Neg: -2.0

12. Let $f: \mathbf{R} \rightarrow \mathbf{R}$ be a continuous function. If $\int_0^x f(2t) dt = \frac{x}{\pi} \sin(\pi x)$ for all $x \in \mathbf{R}$, then $f(2)$ is equal to

- (A) -1
- (B) 0
- (C) 1
- (D) 2

Question 13 (JAM 2007 • Q13 • ID: JAM_2007_Q13)

MCQ

+6 M

Neg: -2.0

13. Let $\vec{u} = (ae^x \sin y - 4x)\hat{i} + (2y + e^x \cos y)\hat{j} + az\hat{k}$, where a is a constant. If the line integral $\oint_C \vec{u} \cdot d\vec{r}$ over every closed curve C is zero, then a is equal to

- (A) -2
- (B) -1
- (C) 0
- (D) 1

Question 14 (JAM 2007 • Q14 • ID: JAM_2007_Q14)

MCQ

+6 M

Neg: -2.0

14. One of the integrating factors of the differential equation $(y^2 - 3xy)dx + (x^2 - xy)dy = 0$ is

- (A) $1/(x^2 y^2)$
- (B) $1/(x^2 y)$
- (C) $1/(x y^2)$
- (D) $1/(x y)$

Question 15 (JAM 2007 • Q15 • ID: JAM_2007_Q15)

MCQ

+6 M

Neg: -2.0

15. Let C denote the boundary of the semi-circular disk $D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1, y \geq 0\}$ and let $\varphi(x, y) = x^2 + y$ for $(x, y) \in D$. If $\hat{n}$ is the outward unit normal to C , then the integral $\oint_C (\vec{\nabla} \varphi) \cdot \hat{n} ds$, evaluated counter-clockwise over C , is equal to

- (A) 0
- (B) $\pi - 2$
- (C) π
- (D) $\pi + 2$

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

IIT JAM 2007 Official Paper • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+6	A
2	MCQ	+6	A
3	MCQ	+6	B
4	MCQ	+6	C
5	MCQ	+6	D
6	MCQ	+6	C
7	MCQ	+6	A
8	MCQ	+6	D

Q. No.	Type	Marks	Official Key
9	MCQ	+6	B
10	MCQ	+6	C
11	MCQ	+6	B
12	MCQ	+6	A
13	MCQ	+6	D
14	MCQ	+6	B
15	MCQ	+6	C

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.